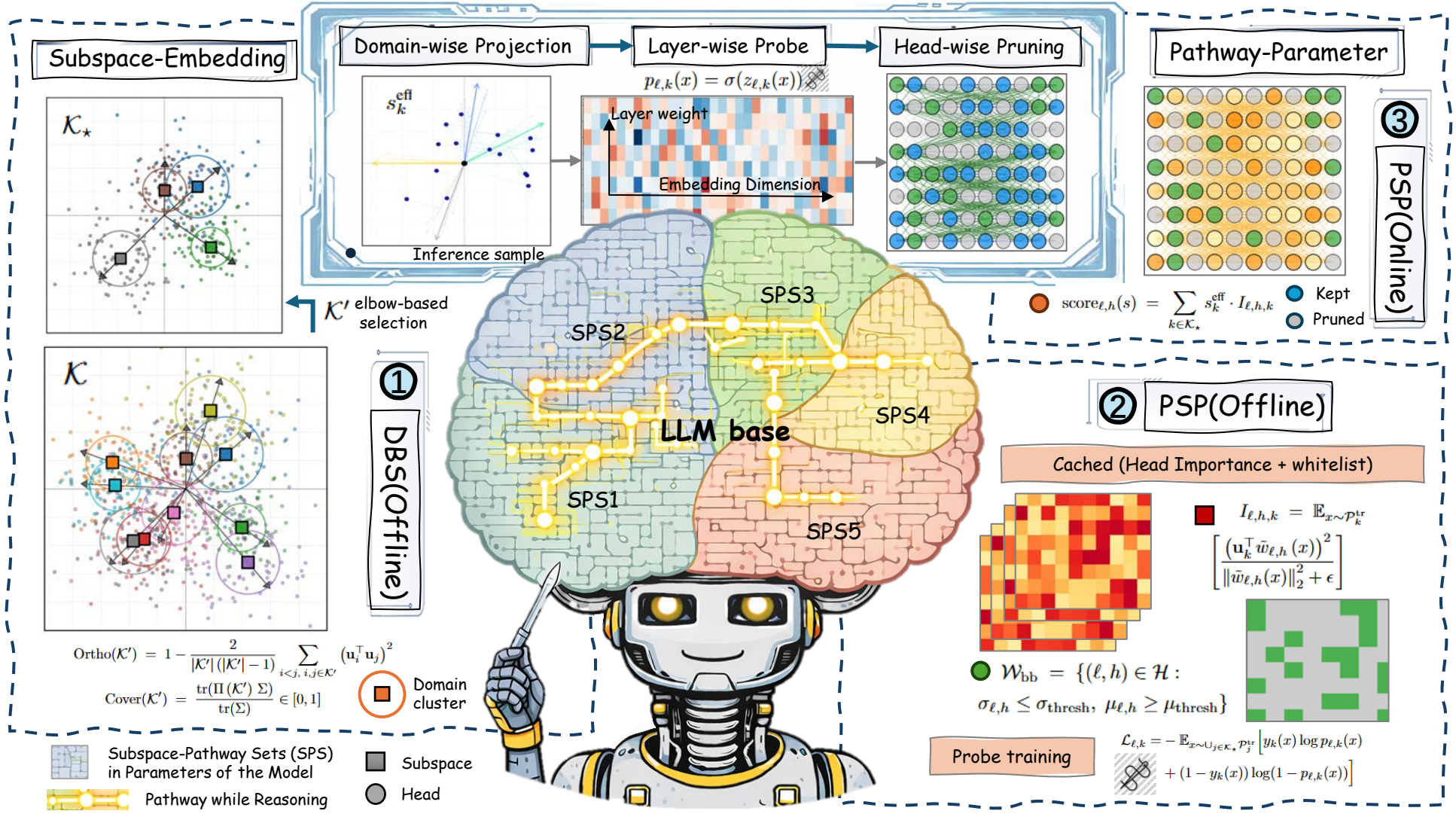
Subspace-Embedding
Domain-wise Projection
Layer-wise Probe
Head-wise Pruning
Pathway-Parameter
s_k^{eff}
Inference sample
$p_{\ell,k}(x) = \sigma(z_{\ell,k}(x))$
Layer weight
Embedding Dimension
$\mathcal{K}_\star$
$\mathcal{K}'$ elbow-based selection
$\mathcal{K}$
① DBS(Offline)
$\text{Ortho}(\mathcal{K}') = 1 - \frac{2}{|\mathcal{K}'|(|\mathcal{K}'|-1)} \sum_{i<j,\, i,j\in\mathcal{K}'} (\mathbf{u}_i^\top \mathbf{u}_j)^2$
$\text{Cover}(\mathcal{K}') = \frac{\text{tr}(\Pi(\mathcal{K}')\,\Sigma)}{\text{tr}(\Sigma)} \in [0,1]$
Domain cluster
SPS1
SPS2
SPS3
SPS4
SPS5
LLM base
$\text{score}_{\ell,h}(s) = \sum_{k\in\mathcal{K}_\star} s_k^{\text{eff}} \cdot I_{\ell,h,k}$
Kept
Pruned
③ PSP(Online)
② PSP(Offline)
Cached (Head Importance + whitelist)
$I_{\ell,h,k} = \mathbb{E}_{x\sim\mathcal{P}_k^{\text{tr}}} \left[\frac{(\mathbf{u}_k^\top \tilde{w}_{\ell,h}(x))^2}{\|\tilde{w}_{\ell,h}(x)\|_2^2 + \epsilon}\right]$
$\mathcal{W}_{\text{bb}} = \{(\ell,h)\in\mathcal{H} : \sigma_{\ell,h} \le \sigma_{\text{thresh}},\ \mu_{\ell,h} \ge \mu_{\text{thresh}}\}$
Probe training
$\mathcal{L}_{\ell,k} = -\mathbb{E}_{x\sim\cup_{j\in\mathcal{K}_\star}\mathcal{P}_j^{\text{tr}}} \left[y_k(x)\log p_{\ell,k}(x) + (1-y_k(x))\log(1-p_{\ell,k}(x))\right]$
Subspace-Pathway Sets (SPS) in Parameters of the Model
Pathway while Reasoning
Subspace
Head